

Performance Testing

Date: 22/09/2026 **Team ID:** 6ab4ccd16f290be8a6b460f4 **Project Name:** aiblognest

Maximum Marks: 5 Marks

Performance testing evaluates how the AI BlogNest API behaves under expected and peak load conditions. This document records the documented testing approach and provides a structured place for measured results.

Step 1: Testing Overview

Field	Details
Testing Tool Used	Postman / Thunder Client collection is documented for API testing. A dedicated performance-load tool was not used.
Type of Testing	Load Testing (planned/documented). Stress and spike testing are not evidenced in the supplied project materials.
Target Module / API	AI BlogNest REST API: authentication, blog CRUD, AI blog generation, and AI summarization endpoints.
Test Environment	Local development environment; documented configuration uses port 5000.
Test Date	22/09/2026

Step 2: Test Scenarios

S.No	Test Scenario / Description	Virtual Users	Duration (sec)	Expected Outcome
1	GET /api/blogs – retrieve all blogs under concurrent read load	10	60	Responses remain within the target average response time.
2	POST /api/blogs – create blogs under concurrent write load	10	60	Successful requests with stable response times.
3	POST /api/ai/generate-blog – AI blog generation under concurrent load	5	60	AI requests complete successfully without errors.
4	POST /api/ai/summarize – summarization under concurrent load	5	60	Summarization requests complete successfully.

Step 3: Performance Test Results

S.No	Metric	Target Value	Actual Value	Status	Remarks
1	Response Time (Avg)	< 2 seconds	Not measured	Pending	No measured load-test result was supplied.
2	Response Time (Max)	< 5 seconds	Not measured	Pending	No measured load-test result was supplied.
3	Throughput (Req/sec)	Record measured value	Not measured	Pending	Requires execution with a performance-testing tool.
4	Error Rate	< 1%	Not measured	Pending	Requires measured request/error counts.
5	CPU Utilization	< 80%	Not measured	Pending	Requires monitoring during the load test.
6	Memory Utilization	< 80%	Not measured	Pending	Requires monitoring during the load test.

Step 4: Observations & Analysis

The supplied project documentation confirms a Node.js/Express REST API with MongoDB/Mongoose, JWT authentication, blog CRUD operations, and Gemini-powered generation and summarization. The supplied Postman/Thunder Client collection documents functional API requests. However, no executed performance-test measurements, monitoring output, or screenshots were supplied. Therefore, the actual performance metrics are intentionally left as Pending rather than being estimated or fabricated.

Recommended execution: Run the four scenarios against the local API, record average/max latency, throughput, error rate, CPU utilization, and memory utilization, then replace the Pending values with the measured results and attach the tool screenshots.

Step 5: Screenshots / Evidence

No performance-tool screenshots were supplied with the source material. Attach the JMeter, Locust, k6, or equivalent performance report/graphs here after execution.

Evidence checklist

- Load-test summary or report
- Response-time graph
- Throughput graph
- Error-rate result
- CPU and memory monitoring evidence